

Probability

Approaches and Parameter Estimation

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

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$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

$$\log L(\theta) = k \log \theta + (n - k) \log(1 - \theta)$$

$$\hat{\theta}_{MLE} = \frac{k}{n}$$

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$P(\theta|\text{data}) \propto P(\text{data}|\theta)P(\theta)$$

$$\theta|\text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

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$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} P(\theta|\text{data})$$

$$P(x_{\text{new}}|\text{data}) = \int P(x_{\text{new}}|\theta)P(\theta|\text{data})d\theta$$

